

EFFECTS OF SPATIAL ABILITIES AND DOMAIN ON ESTIMATION OF PEARSON'S CORRELATION COEFFICIENT

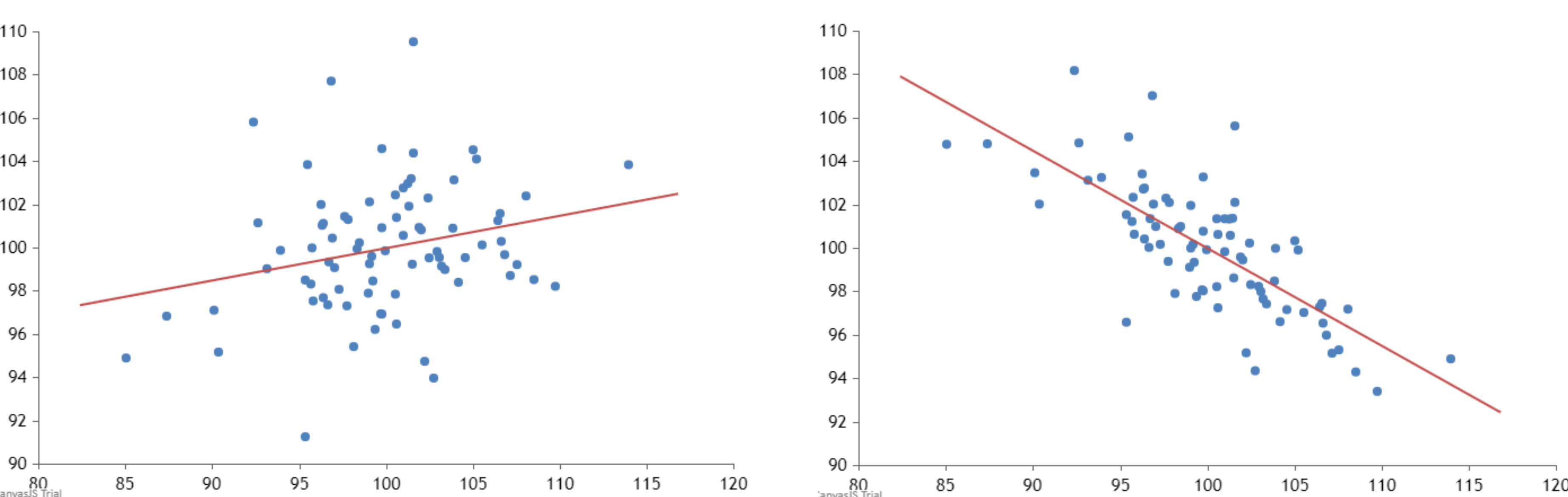
Sara Tandon
Alfie Abdul-Rahman
Rita Borgo



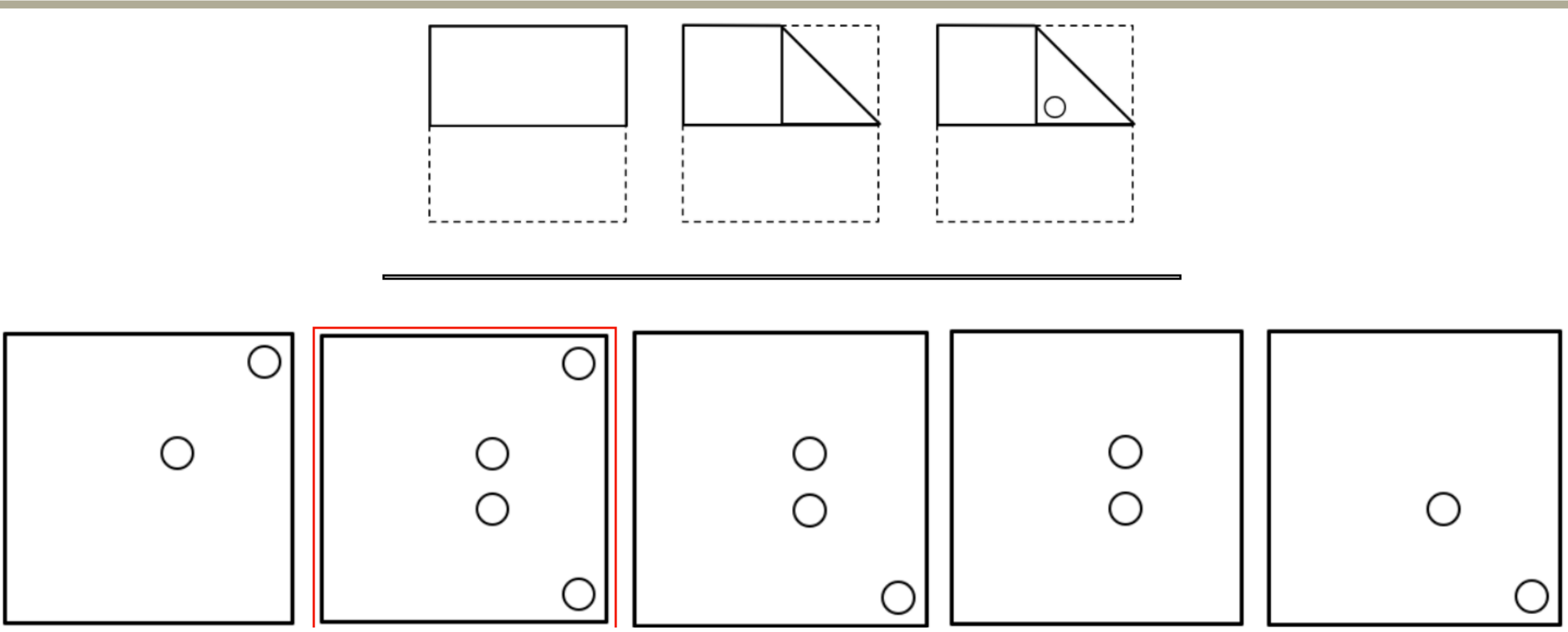
sktandon.github.io
@sarakoyel

ABSTRACT

This work builds on research bridging spatial visualization, psychology, and information visualization to holistically inform visualization design. We expand on exploring the effects of chosen discipline in combination with cognitive abilities on visual tasks through estimation of Pearson's correlation coefficient (PCC). We measure spatial visualization (SV) and visual task performance between psychologists and math & computer scientists (MCS).



SPATIAL VISUALIZATION



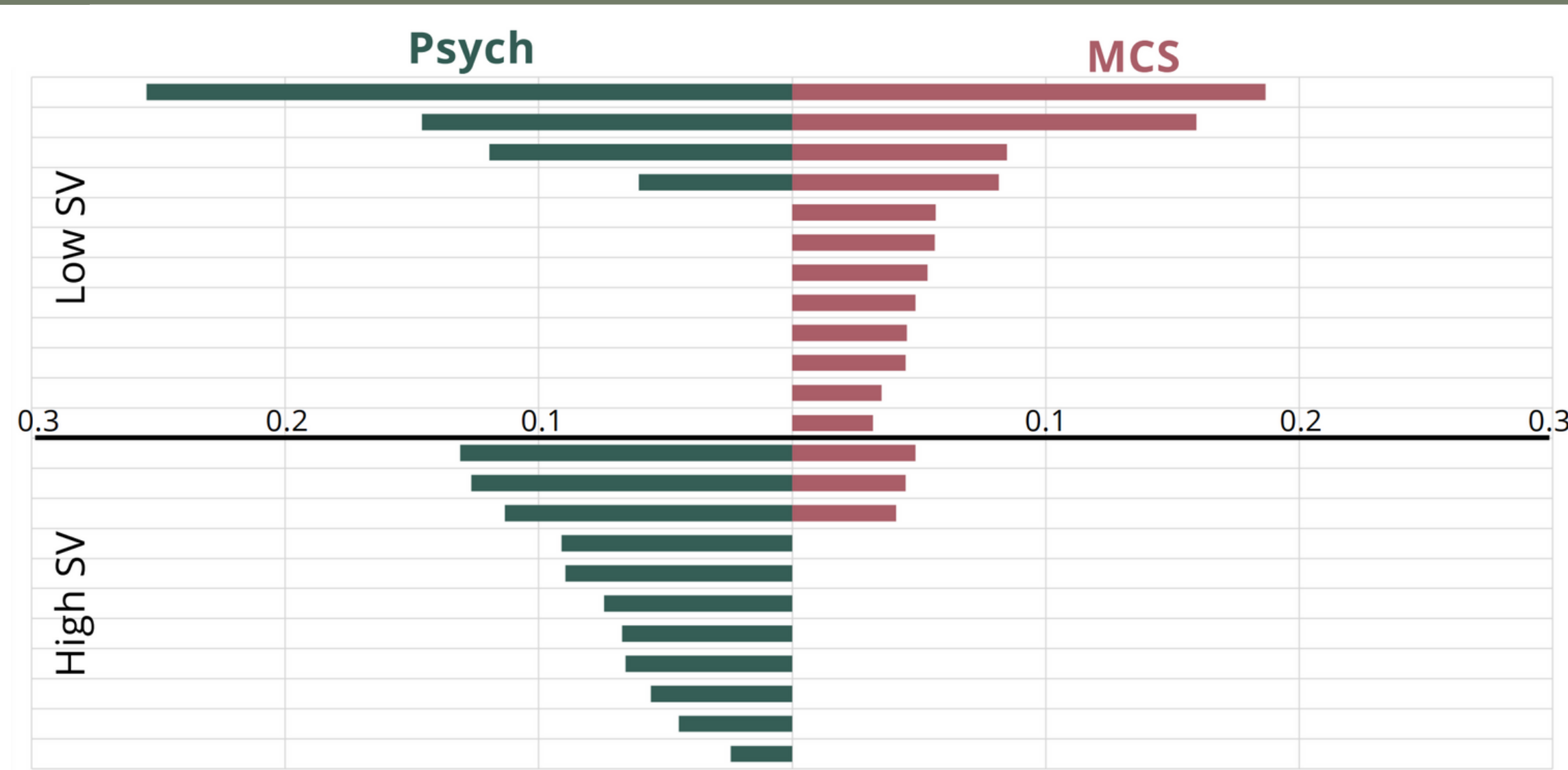
Spatial visualization involves cognitive abilities of retention, manipulation, and rotation of visual images

HYPOTHESES

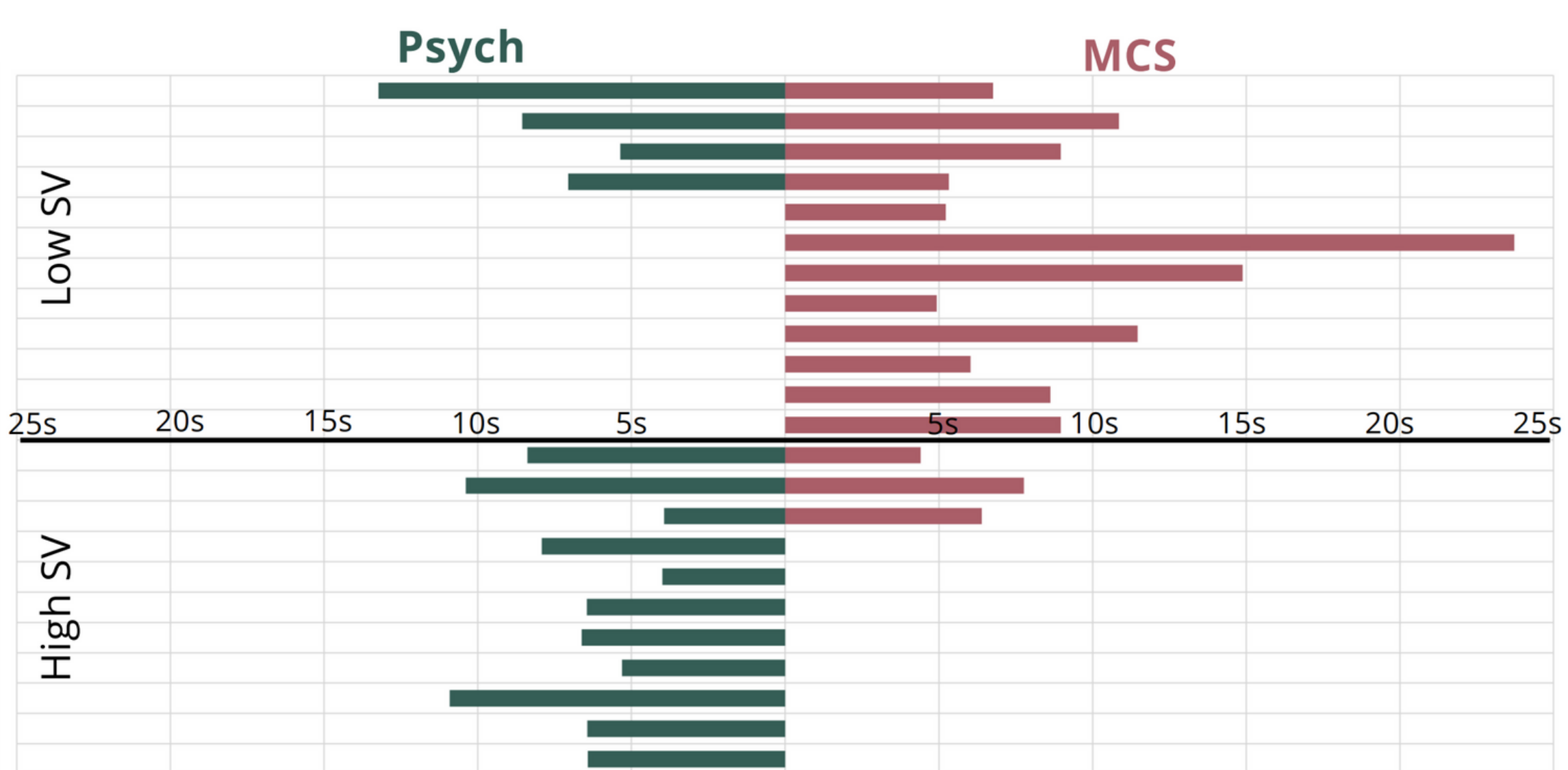
We present the first research into how spatial visualization and domain expertise might affect estimation of PCC in a quantitatively significant way by investigating the following:
H1: SV will differ across domains.
H2: Accuracy and response time (RT) will correlate with SV
H3: Accuracy and RT will vary between disciplines.

METHODS

Part 1: Psychometric spatial visualization assessment.
Part 2: Participants use a slider to manipulate the scatter plot such that the plot reflects their best estimation of a given target PCC between -1 and 1 in 0.1 increments. We utilized randomized data of 80 points with a mean of 100 for all coordinates, standard deviations of 3.0 for the x coordinates and 5.0 for the y coordinates. We chose an estimation task to assess how close each response is to the true Pearson value at different magnitudes.



Individual Mean Differences: split by domain and SV level



Individual RTs: split by domain and SV level, same participant across rows

RESULTS

H1: We confirmed that SV varies with domain, however not what we expected based on past research. Psychometric research demonstrates MCS has increased SV compared to those in social sciences and our previous research demonstrates increased performance on visual tasks by MCS with high SV; our sample gave us insight into performance when abilities do not line up with domain expectations.
H2: We had 16 participants with low SV and 14 with high SV. We detected limited evidence ($p = 0.08$) that those with High SV performed slightly better than Low SV in estimation. We did detect a significant difference ($p < 0.001$) that High SV mean RTs were 2.6s faster than Low SV.
H3: When split by SV level and domain, low and high SV MCS significantly outperformed their Psych counterparts in accuracy while maintaining similar RTs. This indicates that domain expertise in MCS might contribute to elevated levels of estimation of PCC. Further, we see similar accuracy and faster RTs of high SV Psych compared to low SV MCS: indicating SV might affect performance when domain expertise is lacking.

	Psych	MCS	Totals
Low Spatial	3.33 ± 0.5	3.41 ± 0.6	3.4 ± 0.5
High Spatial	7.82 ± 0.6	6.67 ± 1.1	7.57 ± 0.6
Totals	6.86 ± 0.9	4.07 ± 0.8	

SV scores (out of 10) split by domains. Significant differences denoted by red sign across columns and rows. ($\pm$ CI)

		Psych	MCS	Totals
Low Spatial	RT	$8.55s \pm 6.4$	$9.59s \pm 4.5$	$9.39s \pm 3.8$
	Diff	0.145 ± 0.19	0.074 ± 0.04	0.088 ± 0.05
	n	n=4	n=12	
High Spatial	RT	$6.98s \pm 2.5$	$6.19s \pm 3.7$	$6.61s \pm 2.1$
	Diff	0.08 ± 0.05	0.045 ± 0.03	0.072 ± 0.04
	n	n=11	n=3	
Totals	RT	$7.32s \pm 2.4$	$8.91s \pm 3.7$	
	Diff	0.094 ± 0.06	0.068 ± 0.03	

Performance (RT and mean differences) of domains split by SV. Significant differences denoted by red sign. ($\pm$ CI)